High school · counting

Paths, combinations and Pascal

Lesson and method

Before you begin

Products, factorials and unordered choices.

Ideas and method

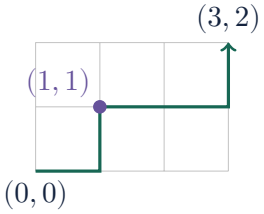
A path from (0,0) to (a,b), using only rightward or upward steps, has $a+b$ steps. Choosing the positions of the a horizontal steps gives $C(a+b,a)$ paths.

For $0 \leq k \leq n, C(n,k) = n!/[k!(n-k)!]$. The order of the chosen objects does not matter. Symmetry $C(n,k) = C(n,n-k)$ corresponds to choosing included or excluded objects.

Pascal's identity $C(n,k) = C(n-1,k-1) + C(n-1,k)$ splits choices by whether one fixed object is included. Paths can also be classified by their last step.

Worked example

From (0,0) to (3,2), there are $C(5,3) = 10$ paths. To pass through (1,1), choose one of two paths to that point and one of three paths onward: $2 \times 3 = 6$ paths.



Explore

Draw a small grid and write the number of arriving paths at each vertex. Add the values of its left and lower neighbours.